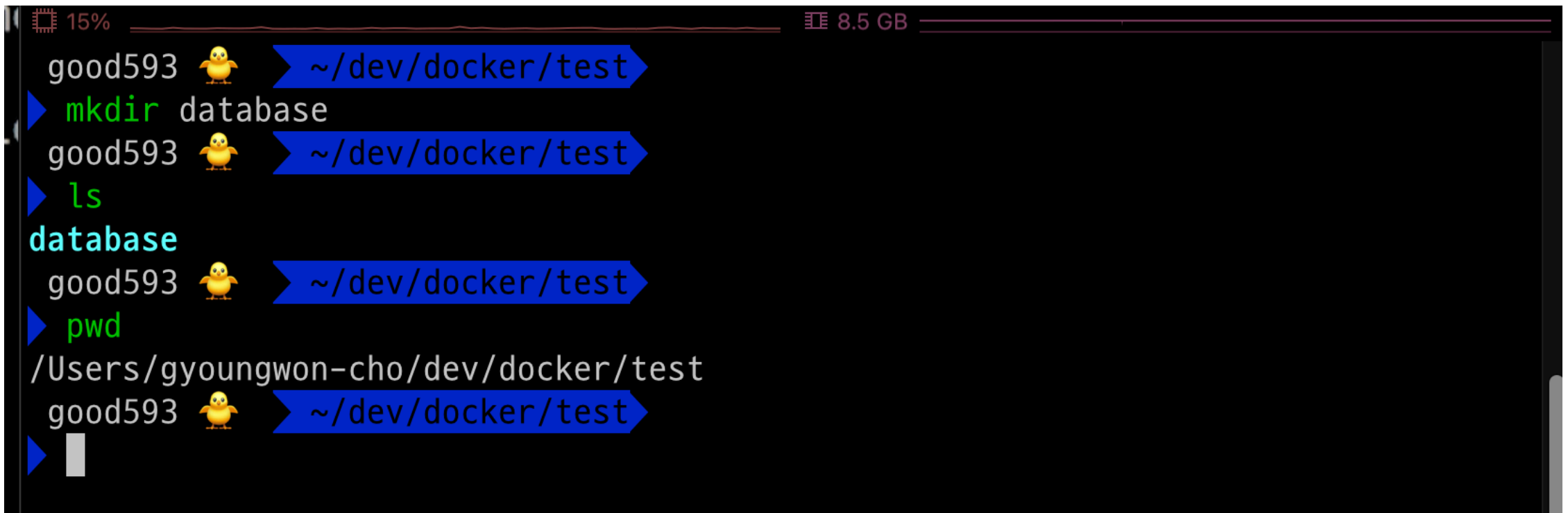


단계1: database 디렉토리 생성

1. 테스트할 폴더로 이동
2. database 디렉토리(폴더) 생성

A terminal window with a dark background. At the top, there is a status bar showing a battery icon at 15% and a memory icon at 8.5 GB. The terminal content shows a user named 'good593' with a penguin emoji icon, currently in the directory '~/dev/docker/test'. The user enters the command 'mkdir database' in green. The prompt changes to show the new directory 'database' in cyan. The user then enters 'ls' in green, and the terminal shows the output 'database' in cyan. Next, the user enters 'pwd' in green, and the terminal shows the full path '/Users/gyoungwon-cho/dev/docker/test'. Finally, the user enters another command, and the prompt returns to '~/dev/docker/test'.

```
good593 🐧 ➤ ~/dev/docker/test  
➤ mkdir database  
good593 🐧 ➤ ~/dev/docker/test  
➤ ls  
database  
good593 🐧 ➤ ~/dev/docker/test  
➤ pwd  
/Users/gyoungwon-cho/dev/docker/test  
good593 🐧 ➤ ~/dev/docker/test  
➤
```

단계2: mysql 컨테이너 생성

- 명령어: `docker run -d --name <컨테이너이름> -v <database 디렉토리 절대경로>:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=<root 비밀번호> mysql:latest`
- `-v <호스트 OS 디렉토리>:<컨테이너 디렉토리>` : 호스트 OS 디렉토리와 컨테이너 디렉토리를 연결
- `/var/lib/mysql` : mysql database의 데이터가 저장되어 있는 디렉토리

```
docker run -d --name db `
-v /Users/gyoungwon-cho/dev/docker/test/database:/var/lib/mysql `
-e MYSQL_ROOT_PASSWORD=pass mysql:latest
```

30%

8.0 GB

good593   ~/dev/docker/test

 **docker** run -d --name db -v /Users/gyoungwon-cho/dev/docker/test/database:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=pass mysql:latest

Unable to find image 'mysql:latest' locally

latest: Pulling from library/mysql

6988ac25ab22: Pull complete

cd2cec5cf6db: Pull complete

e703d567fd0e: Pull complete

de5675f5f5a6: Pull complete

28476d81f322: Pull complete

ccea770348da: Pull complete

842d004c6831: Pull complete

37f7eaf29320: Pull complete

03498da0acb4: Pull complete

00456c5c10e4: Pull complete

Digest: sha256:d7c20c5ba268c558f4fac62977f8c7125bde0630ff8946b08dde44135ef40df3

Status: Downloaded newer image for mysql:latest

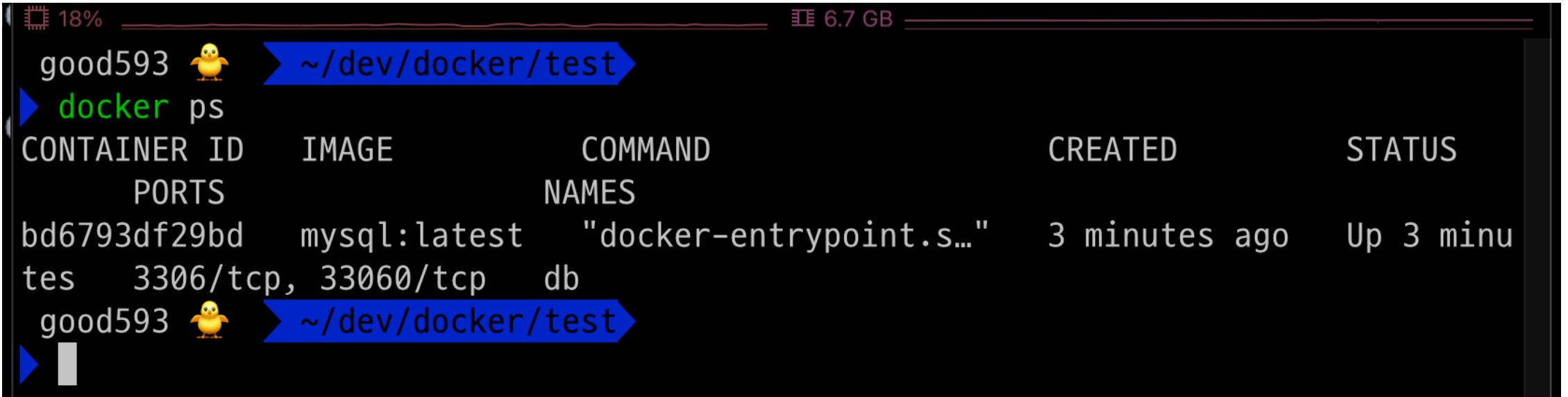
bd6793df29bdefbc11e6e3562d3fcb19960ba7bd0ec1e8f3488f2746ea9c4c10

good593   ~/dev/docker/test

단계3: 작동중인 mysql 컨테이너 확인

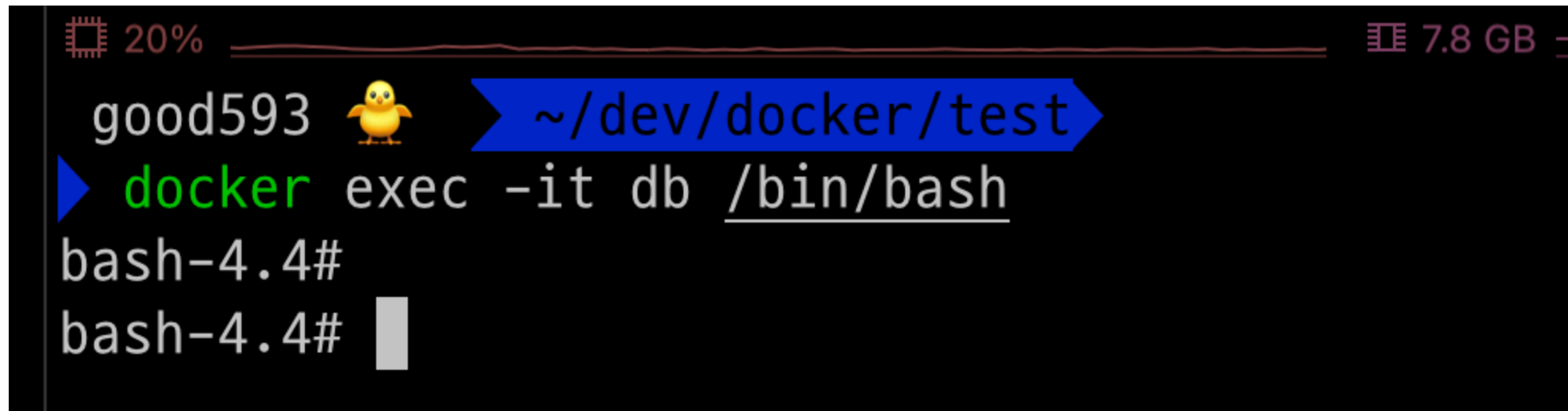
```
docker ps
```



```
good593 🐧 ➤ ~/dev/docker/test  
➤ docker ps  
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS  
PORTS          NAMES  
bd6793df29bd   mysql:latest   "docker-entrypoint.s..." 3 minutes ago  Up 3 minu  
tes           3306/tcp, 33060/tcp  db  
good593 🐧 ➤ ~/dev/docker/test  
➤ █
```

단계4: mysql 컨테이너 접속

```
docker exec -it db /bin/bash
```



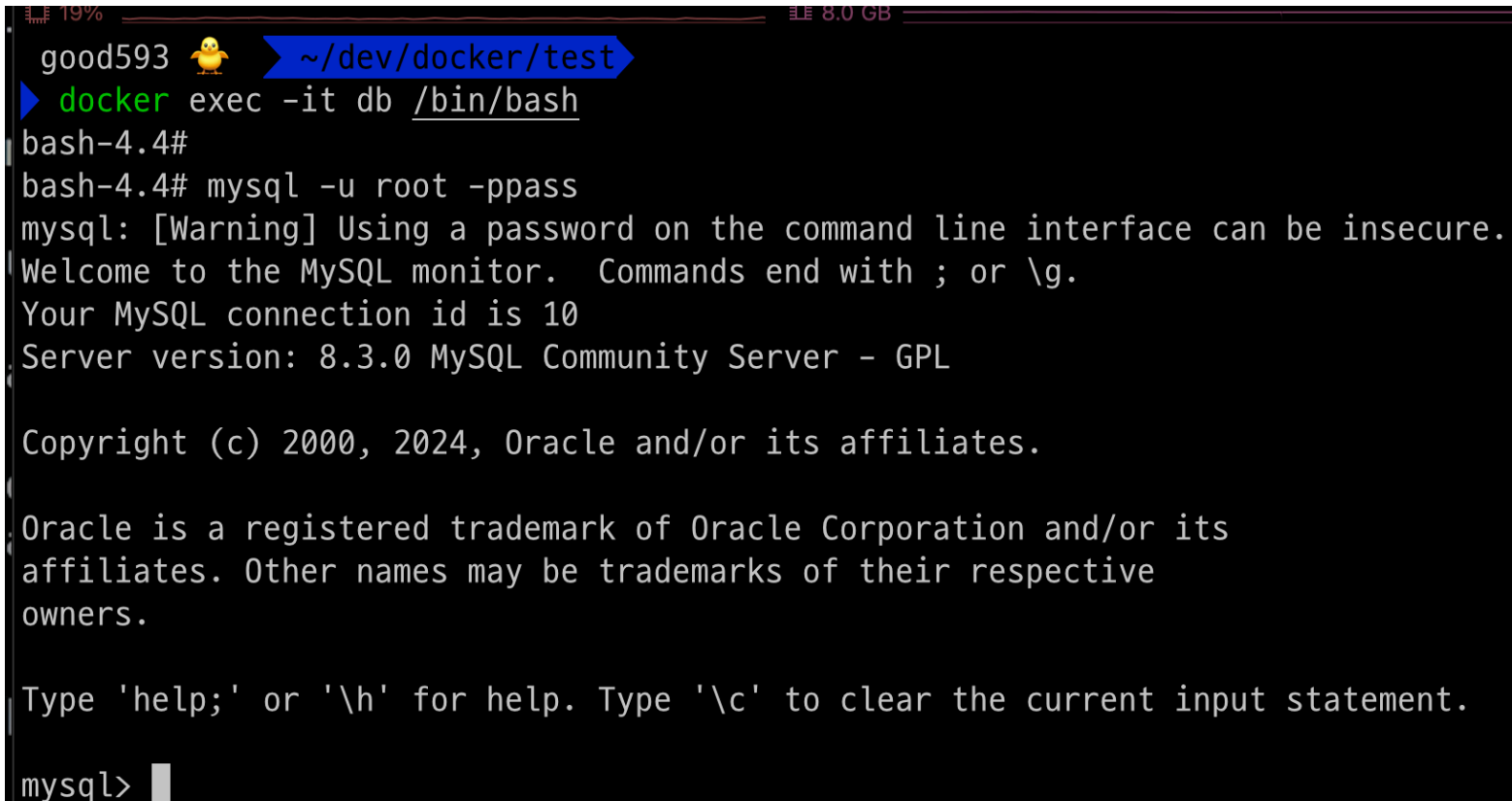
A terminal window with a dark background. At the top, there is a status bar with a CPU icon, '20%', a memory icon, and '7.8 GB'. The user 'good593' is logged in as a penguin user, and the current directory is '~/dev/docker/test'. A blue arrow points to the command 'docker exec -it db /bin/bash'. The command is executed, and the prompt changes from 'good593@~' to 'bash-4.4#'. A second 'bash-4.4#' prompt is shown with a cursor.

```
good593 ~/dev/docker/test  
▶ docker exec -it db /bin/bash  
bash-4.4#  
bash-4.4#
```

단계5: mysql 접속

- 명령어: `mysql -u <계정명> -p<계정비번>`

```
mysql -u root -ppass
```

A terminal window with a dark background. At the top, there's a status bar showing '19%' and '8.0 GB'. The user 'good593' is at the prompt '~ /dev/docker/test'. They run 'docker exec -it db /bin/bash'. The prompt changes to 'bash-4.4#'. Then they run 'mysql -u root -ppass'. The MySQL prompt 'mysql>' appears, followed by a warning message: '[Warning] Using a password on the command line interface can be insecure. Welcome to the MySQL monitor. Commands end with ; or \g. Your MySQL connection id is 10 Server version: 8.3.0 MySQL Community Server - GPL Copyright (c) 2000, 2024, Oracle and/or its affiliates. Oracle is a registered trademark of Oracle Corporation and/or its affiliates. Other names may be trademarks of their respective owners. Type \'help;\' or \'\\h\' for help. Type \'\\c\' to clear the current input statement.' The prompt 'mysql>' is shown at the bottom with a cursor.

```
good593 ~ /dev/docker/test
> docker exec -it db /bin/bash
bash-4.4#
bash-4.4# mysql -u root -ppass
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.3.0 MySQL Community Server - GPL

Copyright (c) 2000, 2024, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\\h' for help. Type '\\c' to clear the current input statement.

mysql>
```

단계6: sql 명령어 실행 및 컨테이너에서 나오기

```
show databases;
```

```
exit # mysql에서 나오기  
exit # mysql 컨테이너에서 나오기
```

```
mysql> show databases;  
+-----+  
| Database                |  
+-----+  
| information_schema      |  
| mysql                   |  
| performance_schema      |  
| sys                     |  
+-----+  
4 rows in set (0.02 sec)
```

```
mysql>  
mysql> exit  
Bye  
bash-4.4#  
bash-4.4# exit  
exit  
good593 🐛 ~/dev/docker/test
```

단계7: database 디렉토리 확인

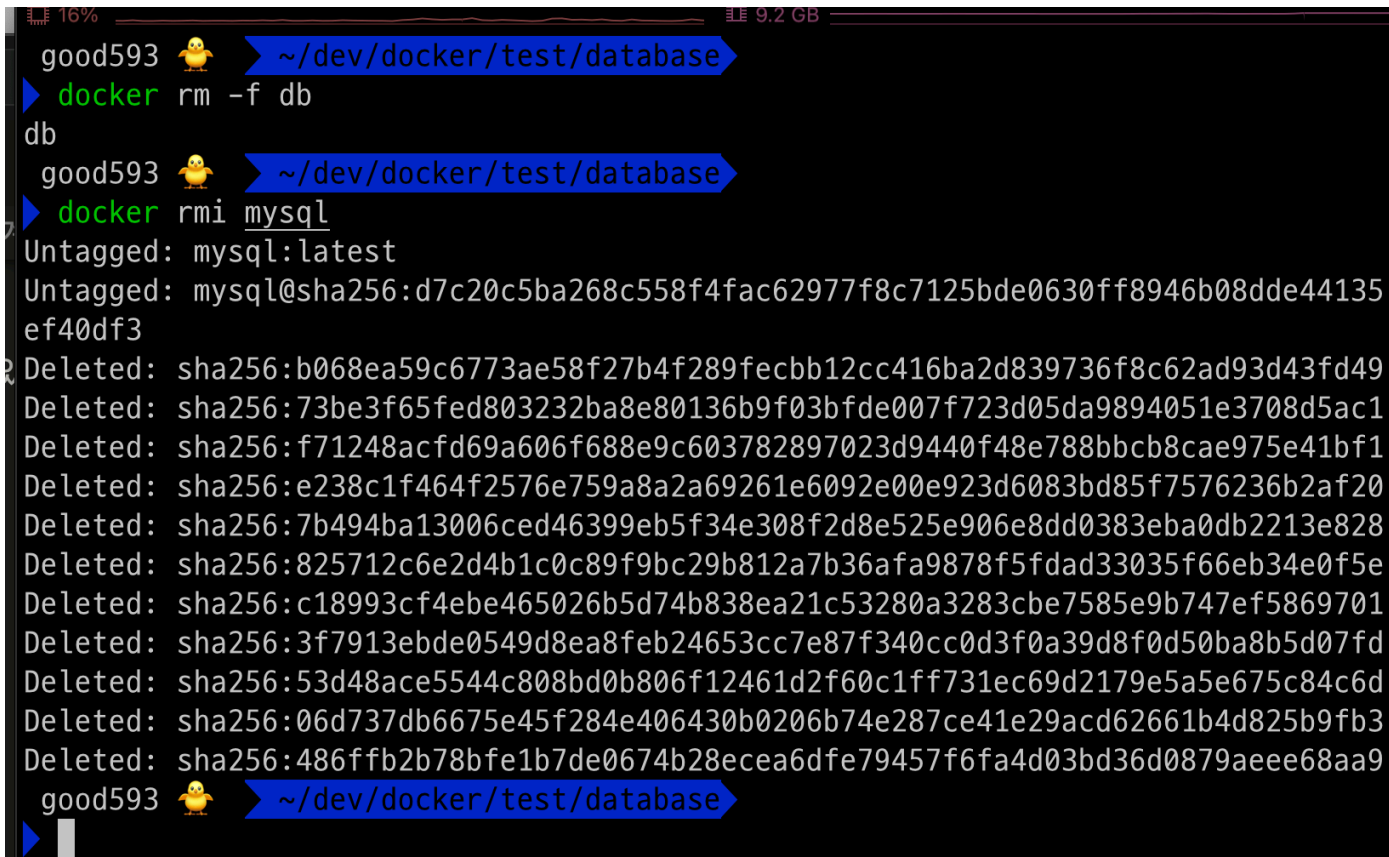
- mysql database에서 사용하는 모든 데이터가 저장되어 있음

```
cd database
ls
```

```
good593 ~/dev/docker/test
> ls
database
good593 ~/dev/docker/test
> cd database
good593 ~/dev/docker/test/database
> ls
#ib_16384_0.dblwr  binlog.000003  ibdata1  private_key.pem
#ib_16384_1.dblwr  binlog.index   ibtmp1    public_key.pem
#innodb_redo      ca-key.pem     mysql     server-cert.pem
#innodb_temp      ca.pem         mysql.ibd  server-key.pem
auto.cnf          client-cert.pem mysql.sock  sys
binlog.000001     client-key.pem mysql_upgrade_info undo_001
binlog.000002     ib_buffer_pool performance_schema undo_002
good593 ~/dev/docker/test/database
>
```

단계8: mysql 컨테이너 & 이미지 삭제

```
docker rm -f db
docker rmi mysql
```

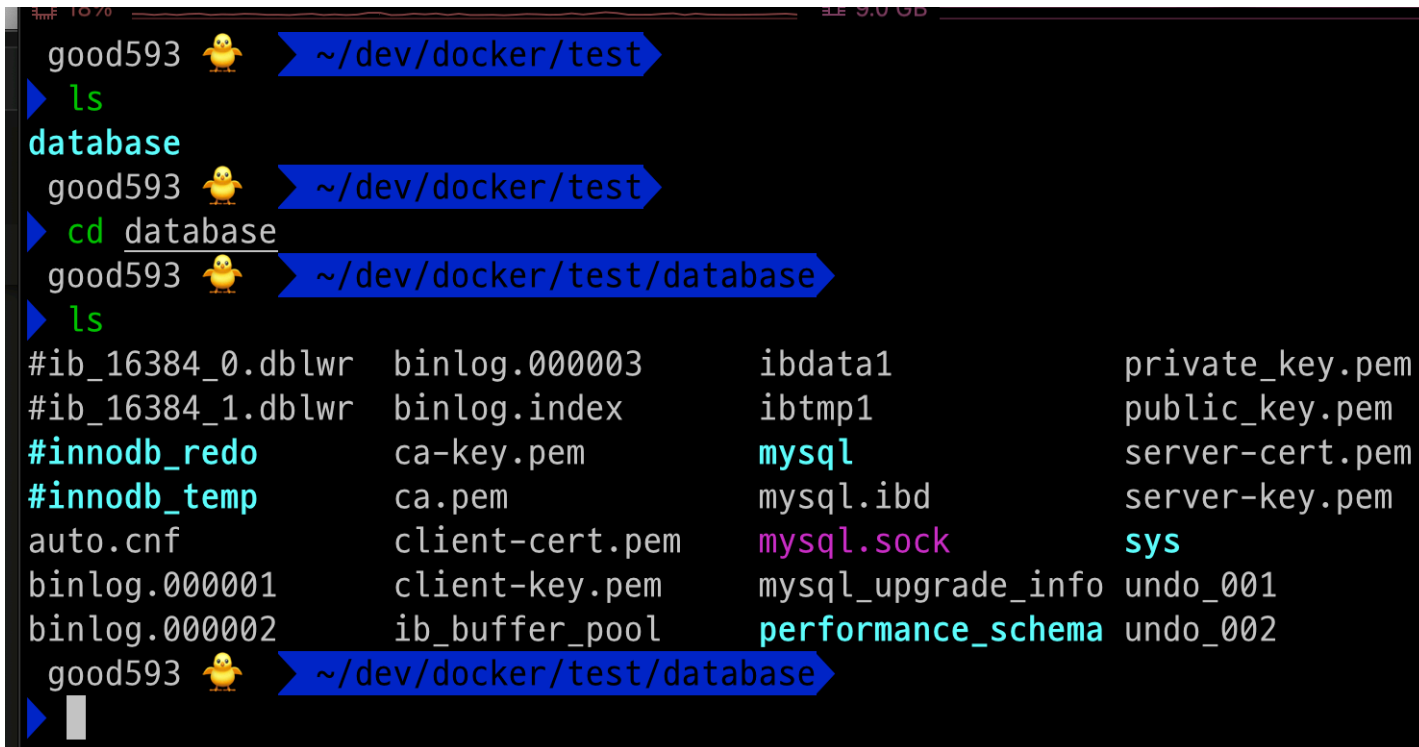
A terminal window with a dark background and light text. The prompt is 'good593' followed by a penguin icon and a blue arrow pointing to the path '~/dev/docker/test/database'. The user enters 'docker rm -f db', and the output is 'db'. Then, the user enters 'docker rmi mysql', and the output shows 'Untagged: mysql:latest' followed by a long list of deleted image IDs (sha256 hashes). The prompt returns to the initial state.

```
good593 ~/dev/docker/test/database
> docker rm -f db
db
good593 ~/dev/docker/test/database
> docker rmi mysql
Untagged: mysql:latest
Untagged: mysql@sha256:d7c20c5ba268c558f4fac62977f8c7125bde0630ff8946b08dde44135ef40df3
Deleted: sha256:b068ea59c6773ae58f27b4f289fecbb12cc416ba2d839736f8c62ad93d43fd49
Deleted: sha256:73be3f65fed803232ba8e80136b9f03bfde007f723d05da9894051e3708d5ac1
Deleted: sha256:f71248acfd69a606f688e9c603782897023d9440f48e788bbcb8cae975e41bf1
Deleted: sha256:e238c1f464f2576e759a8a2a69261e6092e00e923d6083bd85f7576236b2af20
Deleted: sha256:7b494ba13006ced46399eb5f34e308f2d8e525e906e8dd0383eba0db2213e828
Deleted: sha256:825712c6e2d4b1c0c89f9bc29b812a7b36afa9878f5fdad33035f66eb34e0f5e
Deleted: sha256:c18993cf4ebe465026b5d74b838ea21c53280a3283cbe7585e9b747ef5869701
Deleted: sha256:3f7913ebde0549d8ea8feb24653cc7e87f340cc0d3f0a39d8f0d50ba8b5d07fd
Deleted: sha256:53d48ace5544c808bd0b806f12461d2f60c1ff731ec69d2179e5a5e675c84c6d
Deleted: sha256:06d737db6675e45f284e406430b0206b74e287ce41e29acd62661b4d825b9fb3
Deleted: sha256:486ffb2b78bfe1b7de0674b28ecea6dfe79457f6fa4d03bd36d0879ae668aa9
good593 ~/dev/docker/test/database
>
```

단계9: mysql database의 데이터는 삭제되지 않음

- 즉, 컨테이너가 삭제가 되더라도 관련 데이터는 보존할 수 있음

```
cd database  
ls
```



```
good593 ~ /dev/docker/test  
ls  
database  
good593 ~ /dev/docker/test  
cd database  
good593 ~ /dev/docker/test/database  
ls  
#ib_16384_0.dblwr  binlog.000003  ibdata1  private_key.pem  
#ib_16384_1.dblwr  binlog.index  ibtmp1  public_key.pem  
#innodb_redo  ca-key.pem  mysql  server-cert.pem  
#innodb_temp  ca.pem  mysql.ibd  server-key.pem  
auto.cnf  client-cert.pem  mysql.sock  sys  
binlog.000001  client-key.pem  mysql_upgrade_info  undo_001  
binlog.000002  ib_buffer_pool  performance_schema  undo_002  
good593 ~ /dev/docker/test/database
```